Lecture 25 Cluster Analysis II

STAT 441: Statistical Methods for Learning and Data Mining

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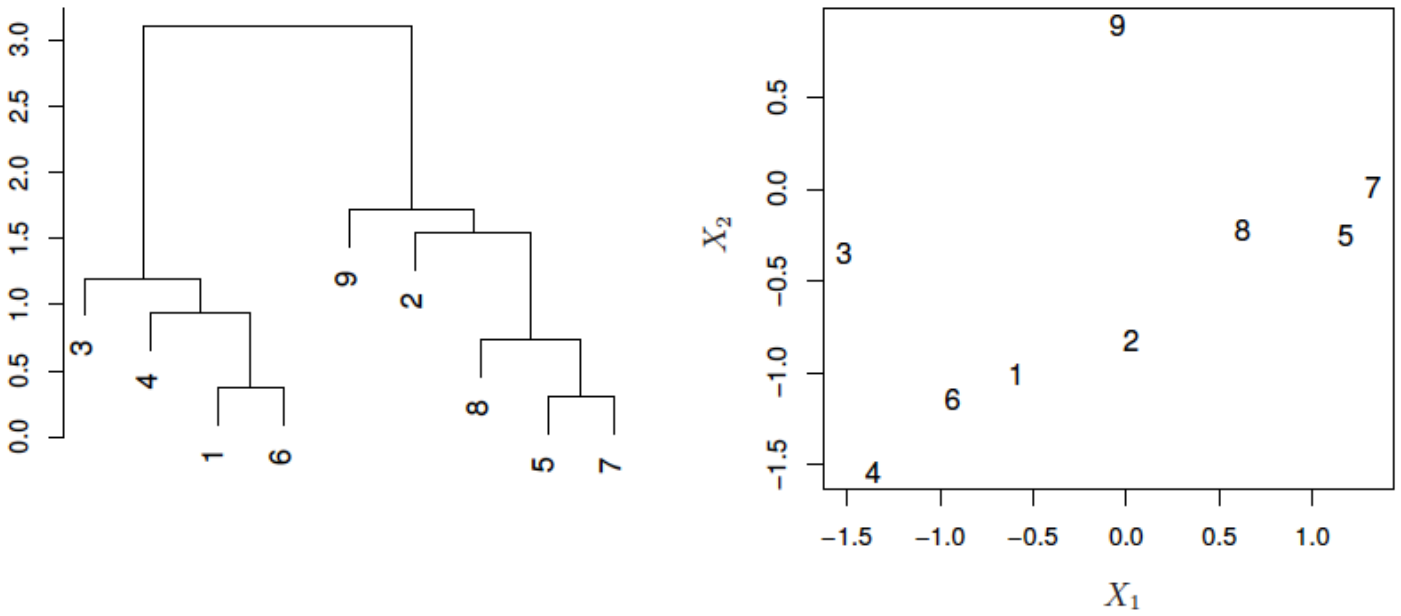
Outline

1. Hierarchically clustering
2. K-means Clustering
3. Other Issues
4. Summary and Remark

Another Example

- An illustration of how to properly interpret a dendrogram with nine observations in two-dimensional space. The raw data on the right was used to generate the dendrogram on the left.



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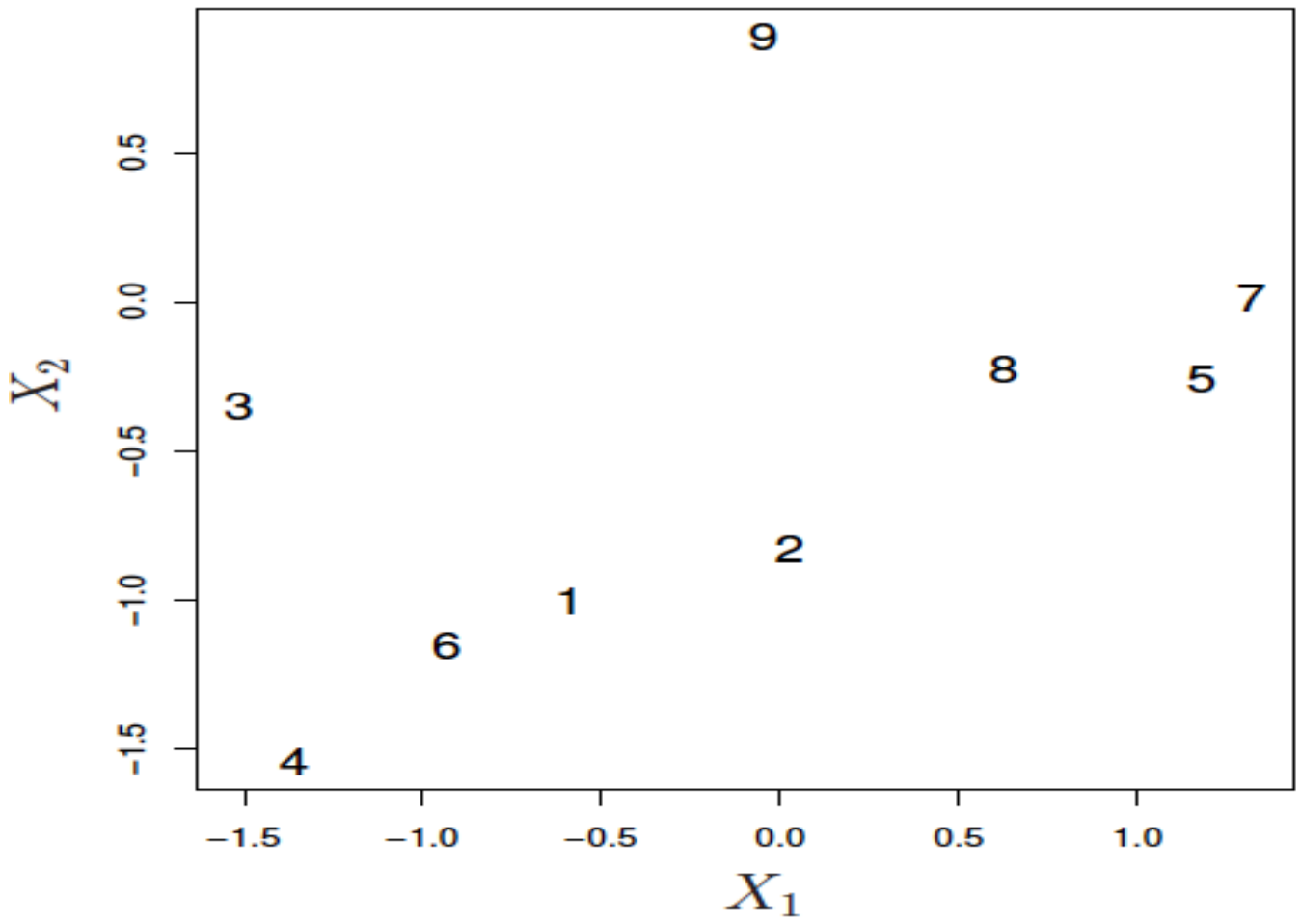
Another Example

- Observations 5 and 7 are quite similar to each other, as are observations 1 and 6.
- However, observation 9 is no more similar to observation 2 than it is to observations 8, 5, and 7, even though observations 9 and 2 are close together in terms of horizontal distance.
- This is because observations 2, 8, 5, and 7 all fuse with observation 9 at the same height, approximately 1.8.

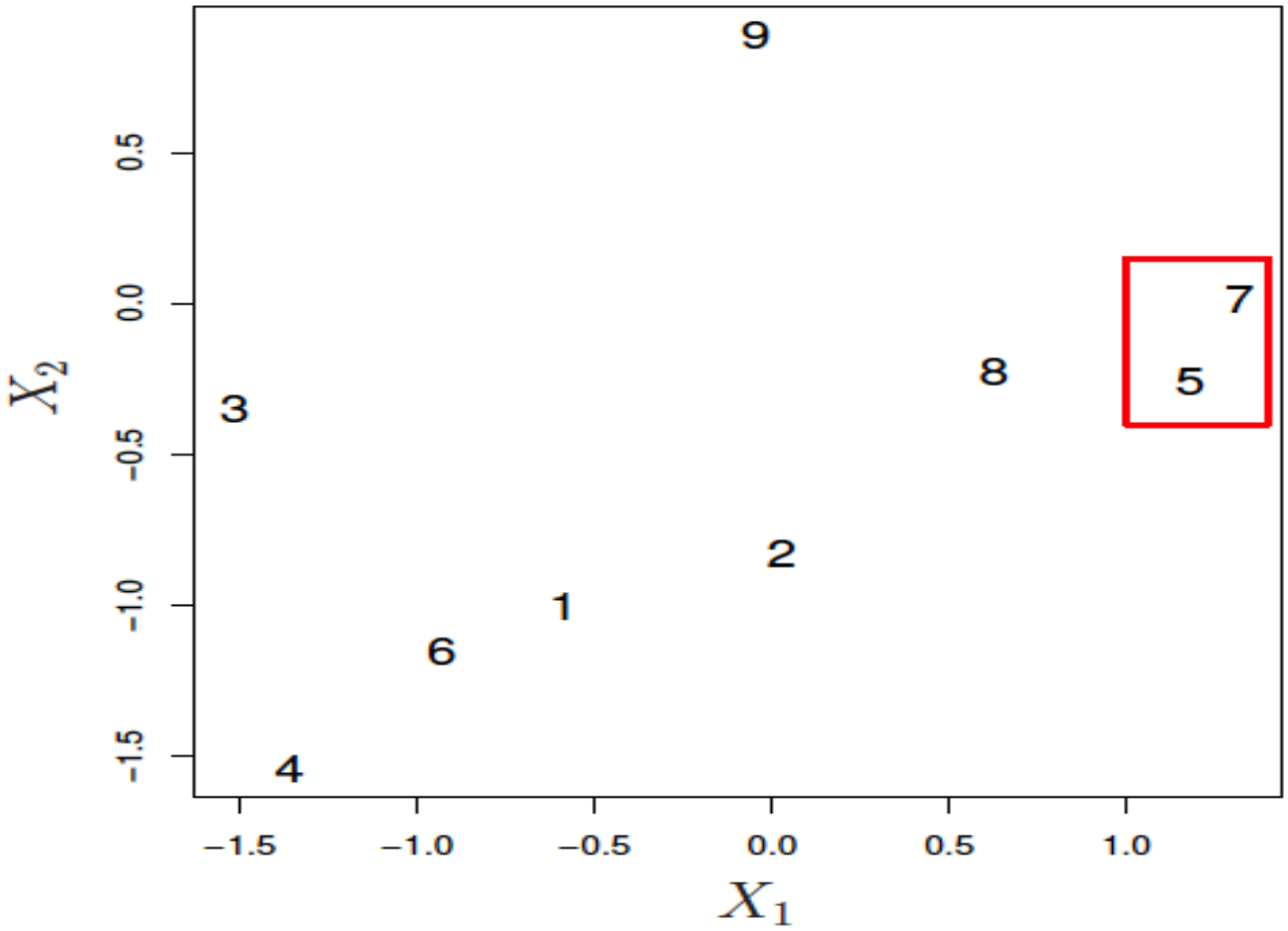
Another Example

Merges in previous example



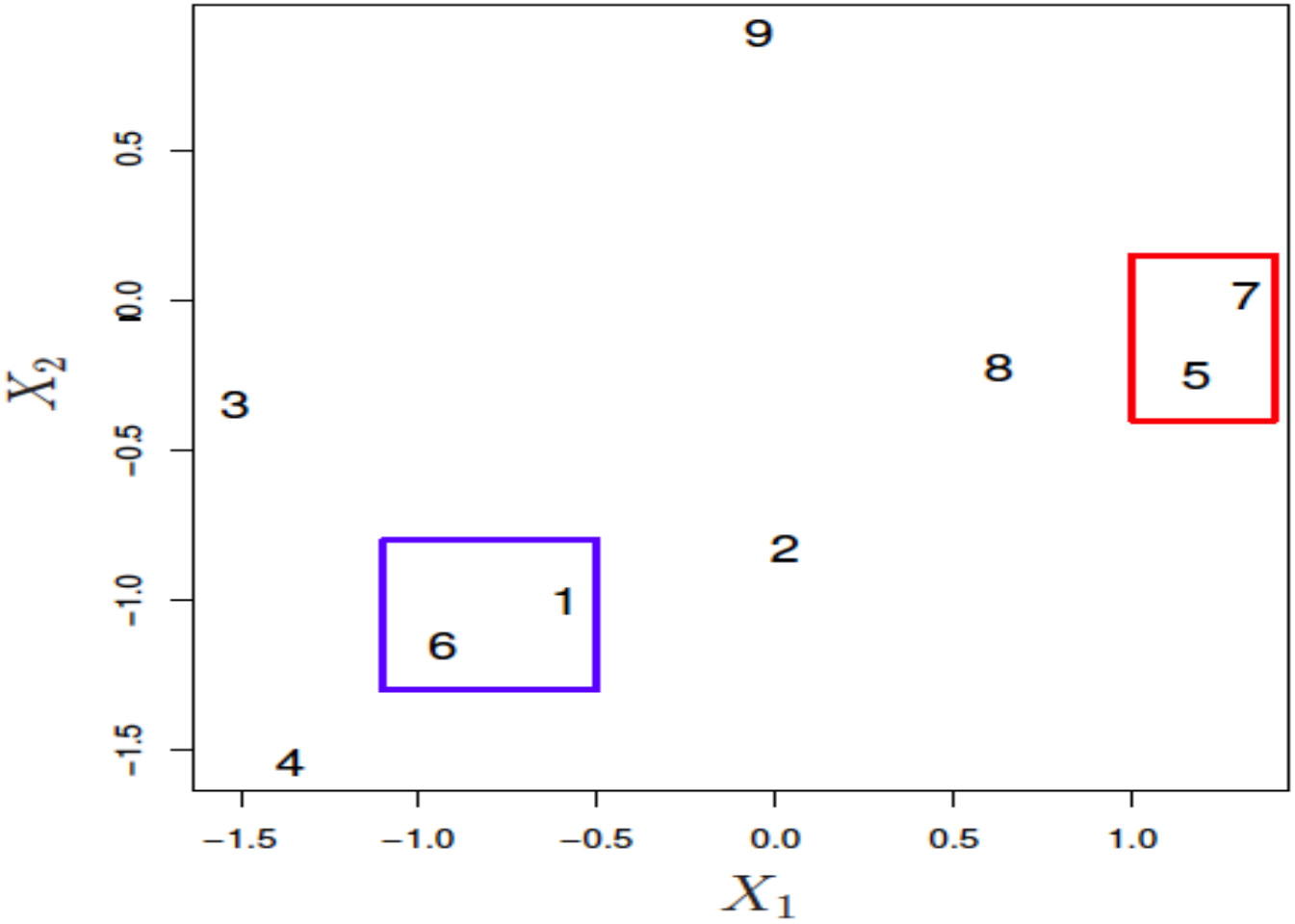
Another Example

Merges in previous example



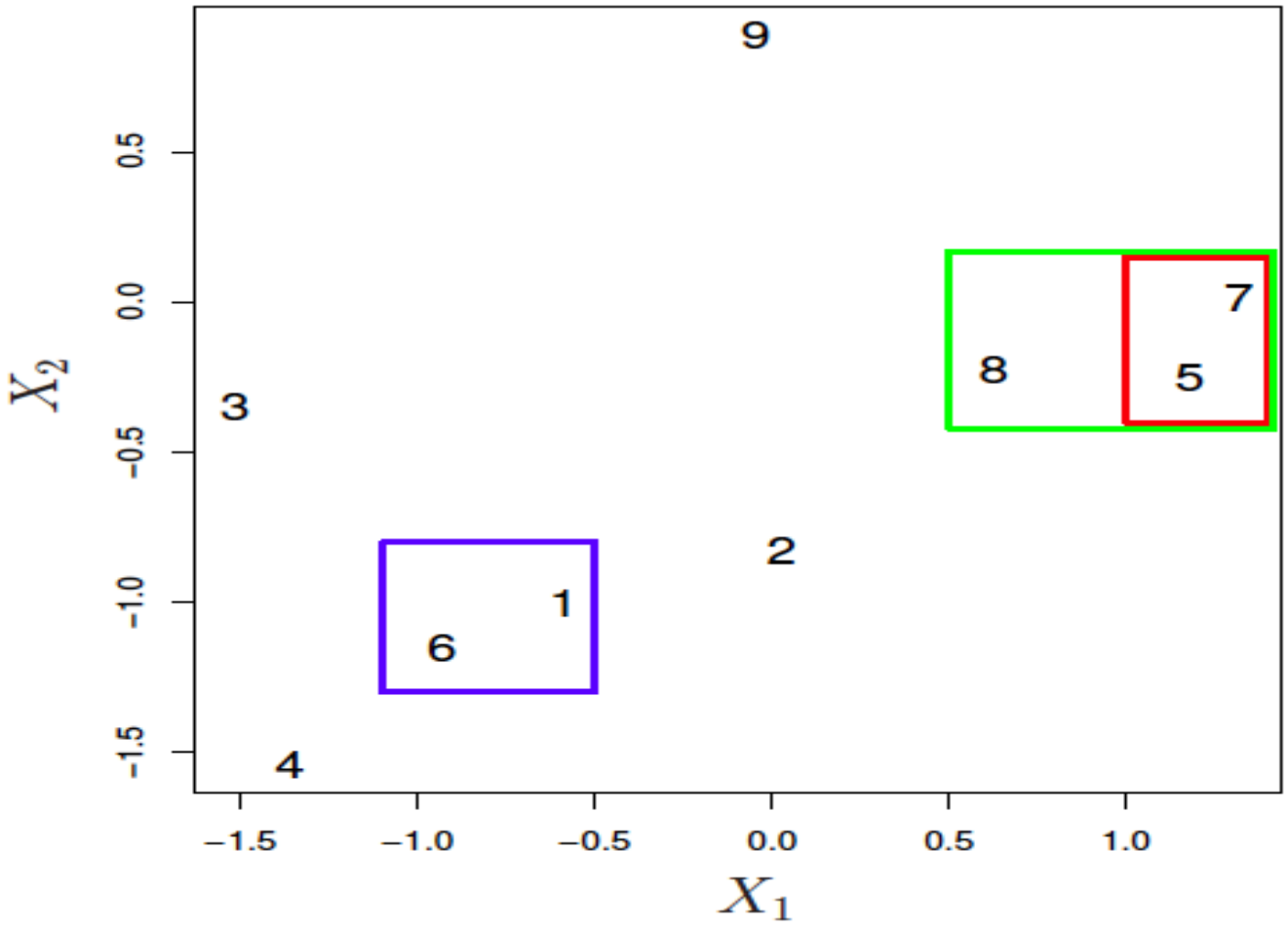
Another Example

Merges in previous example



Another Example

Merges in previous example



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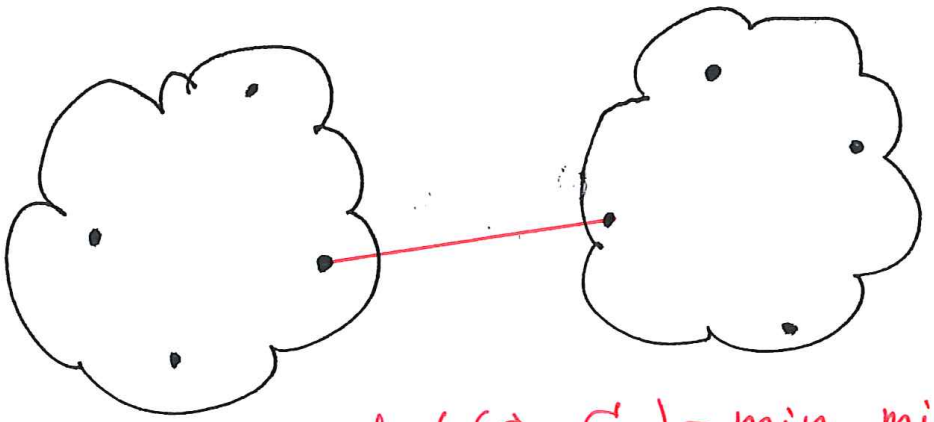
Types of Linkage

- Complete Linkage: Maximal inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the largest of these dissimilarities.
- Single Linkage: Minimal inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the smallest of these dissimilarities.
- Average Linkage: Mean inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the average of these dissimilarities.
- Centroid Linkage: Dissimilarity between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B. Centroid linkage can result in undesirable inversions.

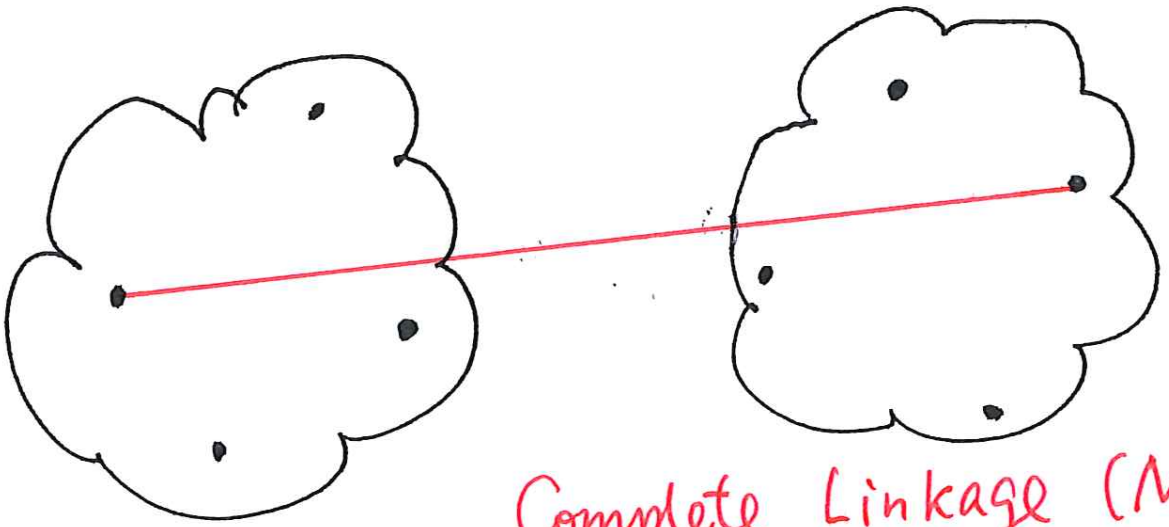
Inter-cluster similarity: single linkage



Single linkage
(MIN)

$$\text{dist}(C_1, C_2) = \min_{x \in C_1} \min_{y \in C_2} \text{dist}(x, y)$$

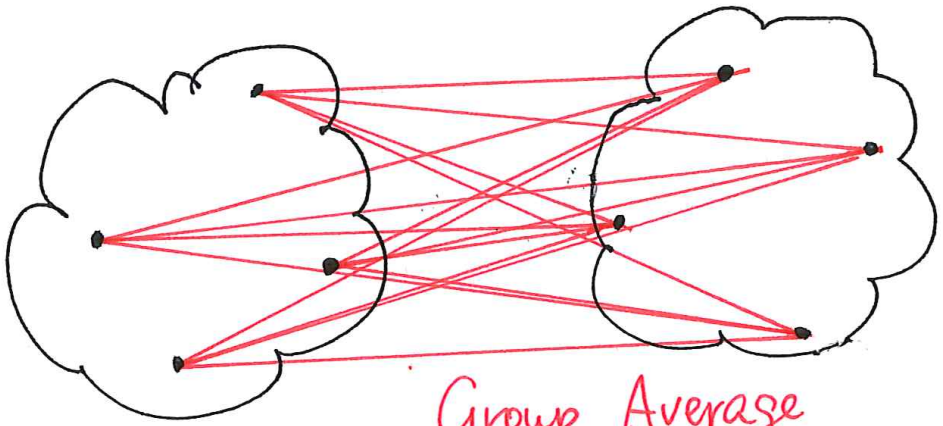
Inter-cluster similarity: complete linkage



Complete Linkage (MAX)

$$\text{dist}(C_1, C_2) = \max_{x \in C_1} \max_{y \in C_2} \text{dist}(x, y)$$

Inter-cluster similarity: group average linkage



Group Average

$$dist(C_1, C_2) = \frac{1}{\#C_1 \times \#C_2} \sum_{x \in C_1} \sum_{y \in C_2} dist(x, y)$$

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Example

With the following distance matrix, generate a cluster tree with single linkage.

	p1	p2	p3	p4
p1	0.00	0.82	0.10	0.35
p2	0.82	0.00	0.91	0.65
p3	0.10	0.91	0.00	0.44
p4	0.35	0.65	0.44	0.00

Solution: Step 1: merge p1, p3: at level 0.1

$C_1 = \{P_1, P_3\}$

$C_2 = \{P_2\}$.

$C_3 = \{P_4\}$

updated proximity matrix:

	C_1	C_2	C_3
C_1	—	0.82	0.35
C_2	0.82	—	0.65
C_3	0.35	0.65	—

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step 2. merge C_1, C_3 at level 0.35.

$C_1 = \{P_1, P_4, P_3\}$

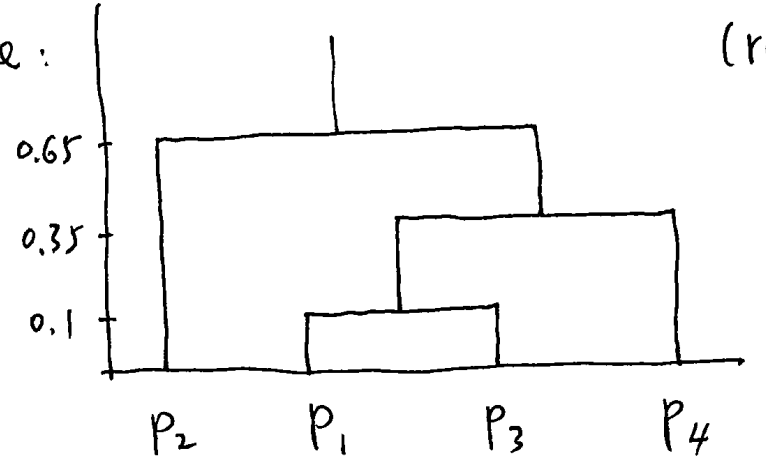
$C_2 = \{P_2\}$.

updated proximity matrix

	C_1	C_2
C_1	—	0.65
C_2	0.65	—

Step 3: merge C_1, C_2 at level 0.65

Cluster tree: (required)



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Ward's Method

- an agglomerative method aimed at finding elliptical clusters (assunming the data are approximately multivariate normal).
- If there are currently K clusters, then let RSS_k be the SS (around the mean) in the kth cluster:

$$RSS_k = \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \bar{\mathbf{x}}_k\|^2, \text{ and } ESS = \sum_k ESS_k.$$

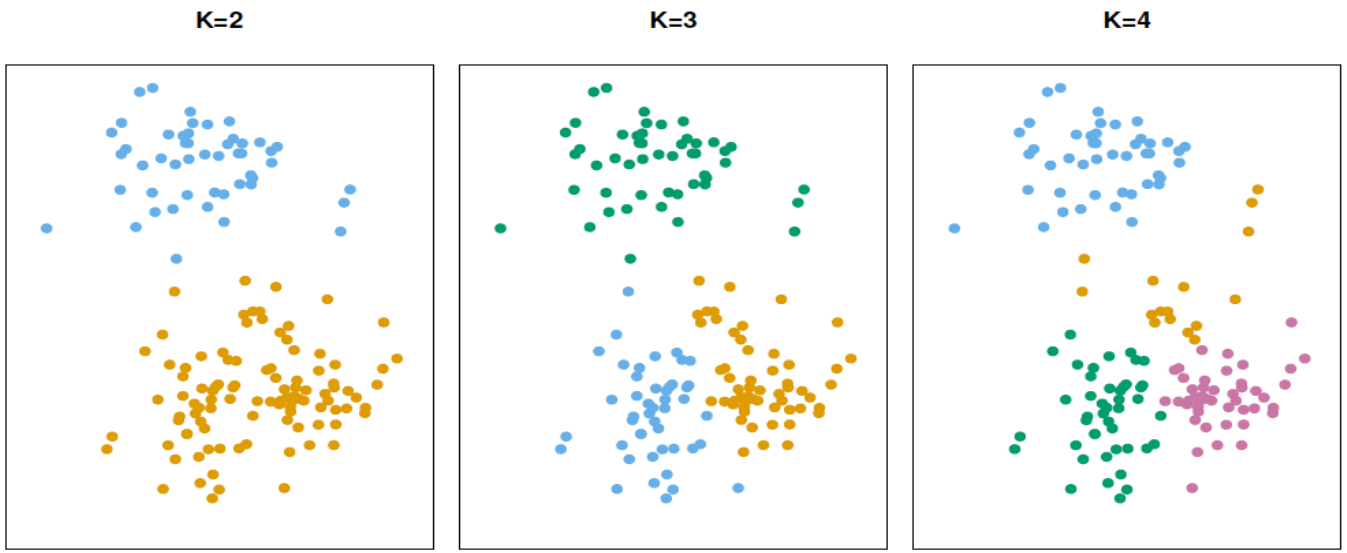
- At each stage, the two clusters to be joined are those that result in the smallest increase in RSS.
- Initially $RSS = 0$ (each observation is its own cluster centroid) (there is only one cluster), eventually

$$RSS = \sum_{i=1}^{n} \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = tr\left[(n-1)\,\mathbf{S}\right].$$

- The user specifies the desired number of clusters, hence the stopping point. This is an optional method in `hclust`.

K-means Clustering

- In K-means clustering, we seek to partition the observations into a pre-specified number of clusters.
- A simulated data set with 150 observations in 2-dimensional space.



K-means Clustering

- Panels show the results of applying K-means clustering with different values of K, the number of clusters.
- The color of each observation indicates the cluster to which it was assigned using the K-means clustering algorithm.
- Note that there is no ordering of the clusters, so the cluster coloring is arbitrary.
- These cluster labels were not used in clustering; instead, they are the outputs of the clustering procedure.

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K-means Clustering

- Let $C_1, \cdots, C_K$ denote sets containing the indices of the observations in each cluster. Theses satisfy two properties:
- 1. $C_1 \cup \cdots \cup C_K = \{1, \cdots, n\}$. In other words, each observation belongs to at least one of the K clusters.
- 2. $C_k \cap C_{k'} = \emptyset$ for all $k \neq k'$. In other words, the clusters are non-overlapping: no observation belongs to more than one cluster.
- For instance, if the ith observation is in the kth cluster, then $i \in C_k$.

K-means Clustering

- The idea behind K-means clustering is that a good clustering is one for which the within-cluster variation is as small as possible.
- The within-cluster variation for cluster C_k is a measure $\mathrm{WCV}(C_k)$ of the amount by which the observations within a cluster differ from each other.
- Hence we want to solve the problem

$$\min_{C_1,\cdots,C_K} \sum_{k=1}^{K} \{WCV(C_k)\}.$$

- In words, this formula says that we want to partition the observations into K clusters such that the total within-cluster variation, summed over all K clusters, is as small as possible.

Within-cluster variation

- Typically we use Euclidean distance

$$WCV(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2,$$

where $|C_k|$ denotes the number of observations in the kth cluster.

- The optimization problem that defines K-means clustering is of the form

$$\min_{C_1,\cdots,C_K} \sum_{k=1}^{K} \left\{ \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 \right\}.$$

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K-Means Clustering Algorithm

- Randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.
- Iterate until the cluster assignments stop changing:
- For each of the K clusters, compute the cluster centroid. The kth cluster centroid is the vector of the p feature means for the observations in the kth cluster.
- Assign each observation to the cluster whose centroid is closest (where closest is defined using Euclidean distance).

K-Means Clustering Algorithm

- This algorithm is guaranteed to decrease the value of the objective at each step. Why? Note that

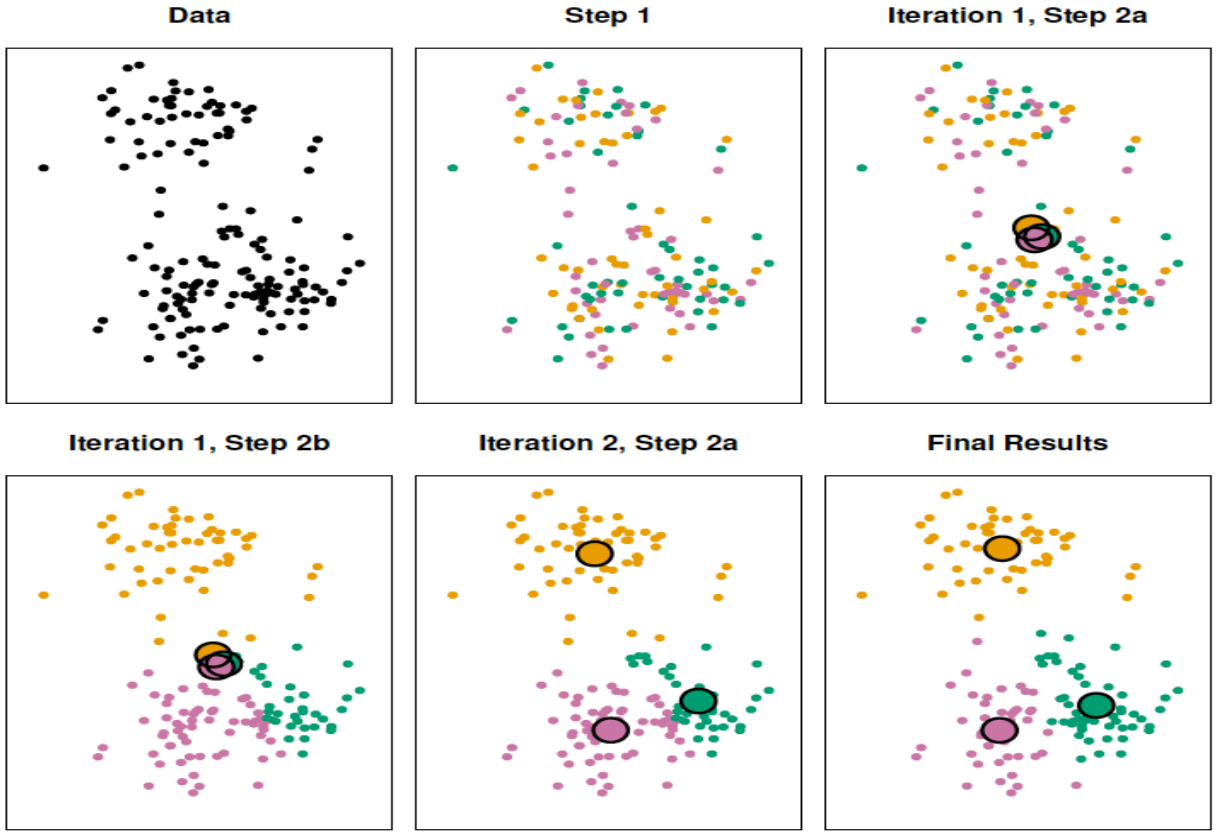
$$\frac{1}{|C_k|}\sum_{i,i'\in C_k}\sum_{j=1}^{p}(x_{ij}-x_{i'j})^2 = 2\sum_{i,\in C_k}\sum_{j=1}^{p}(x_{ij}-\bar{x}_{ij})^2,$$

where $\bar{x}_{kj} = \frac{1}{|C_k|}\sum_{i\in C_k} x_{ij}$ is the mean for feature j in cluster C_k.

- however it is not guaranteed to give the global minimum. Why not?

Example

The progress of the K-means algorithm with $K = 3$ with 10 iterations.



Example

Different starting values and above each plot is the value of the objective. Three different local optima were obtained, one of which resulted in a smaller value of the objective and provides better separation.

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Practical issues

- Should the observations or features first be standardized in some way? For instance, maybe the variables should be centered to have mean zero and scaled to have standard deviation one.
- In the case of hierarchical clustering, What dissimilarity measure should be used? What type of linkage should be used?
- How many clusters to choose? (in both K-means or hierarchical clustering). Difficult problem. No agreed-upon method.

Summary and Remark

- Hierarchical clustering
- K-means clustering
- Read textbook Chapter 10 and `R` code